American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Elementary Division

1. Boolean Algebra

Of the numbers between 20 and 27 inclusive, how many make this Boolean expression TRUE?

$$NOT\ (x \leq 24)\ OR\ (x < 23)$$

A. 0
B. 2
C. 4
D. 6
E. 8

2. Boolean Algebra

Which ordered pair(s) make the following expression TRUE?

$$\sim B\ *\ (A\ +\ B)$$

A. (0,0)
B. (0,1)
C. (1,0)
D. (1,1)
E. None of them

3. Boolean Algebra

Fill in the columns in the following Truth Table. Give the answer as the last column from top to bottom.

A	B	$A * B$	$\sim A$	$\sim A + A * B$	$\sim B$	$(\sim A + A * B) * \sim B$
0	0					
0	1					
1	0					
1	1					

A. 1000
B. 1100
C. 0001
D. 1001
E. 1010

4. Boolean Algebra

Simplify the following Boolean expression:

$$\sim A\ *\ (A\ *\ B\ +\ \sim A\ *\ \sim B)\ +\ \sim B\ *\ (A\ *\ \sim B\ +\ \sim A\ *\ B)$$

A. $\sim A * \sim B$
B. $A * \sim B$
C. $\sim A * B$
D. $\sim A$
E. $\sim B$

5. Boolean Algebra

The Boolean algebra operator $A \oplus B$ is TRUE for the ordered pairs (0,1) and (1,0). Which of the following is equivalent to $\sim(A \oplus B)$?

a. $A * B + \sim A * \sim B$
b. $A * \sim B + \sim A * B$
c. $A + \sim B$
d. $B + \sim A$
e. $\sim A + \sim B + A + B$

A. a
B. b
C. c
D. d
E. e

6. Boolean Algebra

In many elementary schools, teachers either teach Math (M) and Science (S) or Social Studies (SS) and Language Arts (LA). Which Boolean Algebra expression represents this relationship?

A. $M + S * SS + LA$
B. $M * S * SS * LA$
C. $M + S + SS + LA$
D. $M * S + SS * LA$
E. $M * S * \sim(SS * LA)$

American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Elementary Division

1. 布尔代数

在 20 到 27（包括 20 和 27）**之间的**数字中，**请问**有多少个**数字可以使得下列**布尔表达式**的结果**为真？

$$NOT\ (x \leq 24)\ OR\ (x < 23)$$

A. 0
B. 2
C. 4
D. 6
E. 8

2. 布尔代数

请问哪个有序对**可以**使得**下列**表达式**的结果**为真？

$$\sim B\ *\ (A\ +\ B)$$

A. (0,0)
B. (0,1)
C. (1,0)
D. (1,1)
E. 上述答案均不正确

3. 布尔代数

填写以下真值表，将最后一列的内容按照从上到下的顺序依次写出结果。

A	B	$A * B$	$\sim A$	$\sim A + A * B$	$\sim B$	$(\sim A + A * B) * \sim B$
0	0					
0	1					
1	0					
1	1					

A. 1000
B. 1100
C. 0001
D. 1001
E. 1010

4. 布尔代数

请简化下列布尔表达式：

$$\sim A\ *\ (A\ *\ B\ +\ \sim A\ *\ \sim B)\ +\ \sim B\ *\ (A\ *\ \sim B\ +\ \sim A\ *\ B)$$

A. $\sim A\ *\ \sim B$
B. $A\ *\ \sim B$
C. $\sim A\ *\ B$
D. $\sim A$
E. $\sim B$

5. 布尔代数

布尔代数运算符 $A \oplus B$ **对于有序对** (0,1) 和 (1,0) 为真。**请问下列哪个表达式与** ~($A \oplus B$) 等价?

a. $A * B + \sim A * \sim B$
b. $A * \sim B + \sim A * B$
c. $A + \sim B$
d. $B + \sim A$
e. $\sim A + \sim B + A + B$

A. a
B. b
C. c
D. d
E. e

6. 布尔代数

在许多小学里，教师要么教授数学 (M) **和科学** (S)**，要么教授社会研究** (SS) **和语言艺术** (LA)**。请问哪个布尔代数表达式可以表示这种关系？**

A. $M + S * SS + LA$
B. $M * S * SS * LA$
C. $M + S + SS + LA$
D. $M * S + SS * LA$
E. $M * S * \sim(SS * LA)$